Unit 5: Numerical Methods

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from a single Euler step, through the two-slope second-order methods (improved Euler, modified Euler, and Heun), into the four-stage RK4 step, and close with a method comparison and an adaptive-stepping capstone. Throughout, take $\frac{dy}{dt} = f(t, y)$ with step size h. Solutions follow in Part III.

1. Write the Euler update formula, then take one Euler step for $\dfrac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Report y_1 and t_1.
2. Continue from Problem 1 and take a *second* Euler step to find y_2 at $t_2 = 0.4$.
3. State the local and global truncation error orders of Euler's method, and the factor by which the global error changes when h is halved.
4. Take one *improved Euler* step for $\dfrac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Show k_1, k_2, and y_1.
5. Take one *modified Euler* (midpoint) step for the same problem, $\dfrac{dy}{dt} = t + y$, $y(0) = 1$, $h = 0.2$. Show k_1, k_2, and y_1.
6. Write Heun's predictor–corrector formulas and explain in one line why the method is second order.
7. For a general $f(t, y)$, write the four RK4 stage slopes k_1, k_2, k_3, k_4 and the final RK4 update for y_{n+1}.
8. Take one RK4 step for $\dfrac{dy}{dt} = y$ with $y(0) = 1$ and $h = 0.1$. Compute y_1 and compare it to $e^{0.1}$.
9. Rank Euler, improved/modified Euler, Heun, and RK4 by order of accuracy, and give the factor by which each method's global error drops when h is halved.
10. Explain how an adaptive solver uses two estimates of *different* order to decide whether to shrink or grow the step size h.
11. **(Capstone)** Take one RK4 step for $\dfrac{dy}{dt} = t + y$ with $y(0) = 1$ and $h = 0.2$. Compute y_1, then compare it against the Euler result (Problem 1) and the improved Euler result (Problem 4) using the exact solution $y = 2e^t - t - 1$.

Part III — Complete Solutions

Solution 1. The Euler update is $y_{n+1} = y_n + h\,f(t_n, y_n)$. With $f(t, y) = t + y$, the starting slope is $f(0, 1) = 0 + 1 = 1$:

$$y_1 = 1 + 0.2 \cdot 1 = 1.2, \qquad t_1 = 0.2.$$

Solution 2. From $(t_1, y_1) = (0.2,\ 1.2)$ the slope is $f(0.2, 1.2) = 0.2 + 1.2 = 1.4$:

$$y_2 = 1.2 + 0.2 \cdot 1.4 = 1.48, \qquad t_2 = 0.4.$$

Solution 3. A single Euler step drops the $\frac{h^2}{2}y''$ term, so the **local** truncation error is $O(h^2)$. Accumulated over $\sim 1/h$ steps, the **global** error is $O(h)$, making Euler a first-order method. Halving h therefore

$$\text{cuts the global error by about a factor of 2.}$$

Solution 4. Improved Euler uses $k_1 = f(t_n, y_n)$ and $k_2 = f(t_n + h,\ y_n + hk_1)$, then averages:

$$k_1 = f(0, 1) = 1, \qquad k_2 = f(0.2,\ 1 + 0.2 \cdot 1) = f(0.2,\ 1.2) = 1.4.$$

$$y_1 = 1 + \frac{0.2}{2}(1 + 1.4) = 1 + 0.1 \cdot 2.4 = 1.24.$$

Solution 5. Modified Euler reads the slope at the midpoint after a half-step prediction:

$$k_1 = f(0, 1) = 1, \qquad k_2 = f(0.1,\ 1 + 0.1 \cdot 1) = f(0.1,\ 1.1) = 1.2.$$

$$y_1 = 1 + h\,k_2 = 1 + 0.2 \cdot 1.2 = 1.24.$$

Here both second-order methods agree to the digits shown.

Solution 6. Heun predicts with a full Euler step and corrects with the trapezoidal average of the endpoint slopes:

$$\tilde{y}_{n+1} = y_n + hf(t_n, y_n), \qquad y_{n+1} = y_n + \frac{h}{2}\big[f(t_n, y_n) + f(t_{n+1}, \tilde{y}_{n+1})\big].$$

Averaging the two endpoint slopes restores the leading curvature term that plain Euler discards, so the global error is $O(h^2)$ — second order.

Solution 7. The classical RK4 stages and update are

$$k_1 = f(t_n, y_n),$$

$$k_2 = f\left(t_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}k_1\right), \qquad k_3 = f\left(t_n + \tfrac{h}{2},\ y_n + \tfrac{h}{2}k_2\right),$$

$$k_4 = f\left(t_n + h,\ y_n + hk_3\right),$$

$$y_{n+1} = y_n + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right).$$

Solution 8. For $f(t, y) = y$ with $y(0) = 1$, $h = 0.1$:

$$k_1 = 1, \quad k_2 = 1 + 0.05 \cdot 1 = 1.05, \quad k_3 = 1 + 0.05 \cdot 1.05 = 1.0525,$$

$$k_4 = 1 + 0.1 \cdot 1.0525 = 1.10525.$$

$$y_1 = 1 + \frac{0.1}{6}\big(1 + 2(1.05) + 2(1.0525) + 1.10525\big) = 1 + \frac{0.1}{6}(6.31025) = 1.105171.$$

Against $e^{0.1} = 1.1051709\ldots$ the error is below 10^{-6}.

Solution 9. Ranked by order of accuracy:

Euler	1st order $O(h)$	halving $h \Rightarrow \div 2$
Improved / Modified Euler	2nd order $O(h^2)$	halving $h \Rightarrow \div 4$
Heun	2nd order $O(h^2)$	halving $h \Rightarrow \div 4$
RK4	4th order $O(h^4)$	halving $h \Rightarrow \div 16$

Solution 10. An adaptive solver advances each step two ways of *different* order — for example an embedded RK4/RK5 pair, or one full step compared against two half steps. The difference between the two estimates approximates the local error. If that estimate **exceeds** the tolerance, the solver **shrinks** h and retries the step; if it sits comfortably **under** the tolerance, the solver **grows** h for the next step, so the work concentrates where the solution changes fastest.

Solution (Capstone). RK4 for $f(t, y) = t + y$, $y(0) = 1$, $h = 0.2$:

$$k_1 = f(0, 1) = 1, \qquad k_2 = f(0.1,\ 1 + 0.1 \cdot 1) = f(0.1,\ 1.1) = 1.2,$$

$$k_3 = f(0.1,\ 1 + 0.1 \cdot 1.2) = f(0.1,\ 1.12) = 1.22, \qquad k_4 = f(0.2,\ 1 + 0.2 \cdot 1.22) = f(0.2,\ 1.244) = 1.444.$$

$$y_1 = 1 + \frac{0.2}{6}\big(1 + 2(1.2) + 2(1.22) + 1.444\big) = 1 + \frac{0.2}{6}(7.284) = 1.242805.$$

The exact solution gives $y(0.2) = 2e^{0.2} - 0.2 - 1 = 1.242806$, so RK4 is correct to six figures. Compare the one-step estimates:

$$\text{Euler } 1.2 \text{ (err } \approx 4.3 \times 10^{-2}), \quad \text{Improved } 1.24 \text{ (err } \approx 2.8 \times 10^{-3}), \quad \text{RK4 } 1.242805 \text{ (err } \approx 10^{-6}).$$

Unit 5 Key Takeaway

Accuracy is bought with slopes per step. Euler uses one (first order); improved/modified Euler and Heun average two (second order); RK4 weights four as $\frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ (fourth order). On $y' = t + y$ over a single $h = 0.2$ step, the errors fall from $\sim 10^{-2}$ to $\sim 10^{-6}$ across the ladder — and adaptive stepping spends that accuracy budget only where the solution demands it.